Mobile Architectures

Informatics Engineering Department

Coimbra Institute of Engineering

Course context

- 1st degree in Informatics Engineering (daytime and evening classes)
  - Application Development specialization

- Teachers
  - Theoretical classes:
    - Álvaro Santos (ans@isec.pt)
  - Practical classes:
    - Álvaro Santos (ans@isec.pt)
    - Tiago Figueira (tiago@isec.pt)

Objectives

- To understand the functioning of operating systems for mobile devices and their specificities
- To understand the advantages and disadvantages of mobile application development
- To develop native applications for mobile devices
- To develop cross-platform applications

Generic program

- Concepts about mobile systems, disseminated systems and ubiquitous systems
- Concepts on application development for mobile devices
- Mobile application development
  - Components of a mobile application
  - Application lifecycle
  - Definition of screens and visual components
  - User interaction mechanisms
  - Lists
  - Data persistence
  - Performing asynchronous tasks
  - Communication with other devices and Web services
  - Location awareness
  - Sensors
  - Image capture
- Cross-platform application development
  - Advantages and disadvantages of cross-platform development
  - Component-based application development
- Mobile application deployment

Generic program (practical classes)

- Mobile application development
  - Mobile application development guidelines and technologies
  - Android applications in Kotlin
    - XML layouts and Jetpack Compose
  - Cross-platform applications in Flutter
- Other topics
  - Location awareness and maps
  - Web services
  - Firebase
  - …

Workload

- 6 ECTS ⇨ ≈160 hours
  - For the "typical student"

- Contact hours (base: 15 weeks)
  - Theoretical-practical component: 30 hours
  - Practical-laboratory component: 45 hours

- Assessment
  - Written exam: ≈2 hours
  - Defense of works: ≈1 hour

- Hours of unaccompanied study: 80+ hours

Assessment

- The evaluation, on a 0..20 scale, will be carried out through the following components:
  - Written exam (5 marks)
    - Individual written test with limited consultation
    - Minimum mark: 35%
  - Practical works (12 marks)
  - Attendance (3 marks)

Practical work

- Two group works
  - 1st Work (9 marks)
    - Native *Android in Kotlin*
    - Work description will be available in the 1st week of November
    - Delivery date: **2025.01.05 23:59**
  - 2nd Work (3 marks)
    - *Flutter*
    - Work description will be available in the 2nd week of December
    - Delivery date : **2025.01.05 23:59**

Practical work

- Additional delivery notes
  - Submission: *Nónio*
  - ZIP format (***1st student work: uninstall WinRAR***)
    - Technical report in PDF format
    - Project
  - **Penalty: 5% for each hour of delay**

- Mandatory defense
  - Dates: January 6, 7, and 8, 2025

- Minimum mark: 35% in each work

Groups

- Number of elements: **3**
  - Preferably from the same class

- The constitution of the groups must be handed in:
  - *Nónio* (an activity will be available for this purpose)
  - Submitted as a text file with the following format:

```
Elements:
- <full_name_std1>, <std1_id_Nonio>, <practical_class_std1>, <e-mail_isec_std1>
- <full_name_std2>, <std2_id_Nonio>, <practical_class_std2>, <e-mail_isec_std2>
- <full_name_std3>, <std3_id_Nonio>, <practical_class_std3>, <e-mail_isec_std3>
```

- Deadline: 2024.10.31
  - After this date, no more groups will be accepted
    - The work must be carried out individually

- **NOTE: All submissions through *Nónio* must be carried out by only one member of the group, who must associate the remaining members of the group with the delivery**

Attendance

- 3 marks
  - Theoretical classes: 0.1 for each class, up to a maximum of 1
  - Practical classes: 0.2 for each class, up to a maximum of 2

- Erasmus students and students who have a status that exempts them from attending classes (e. g., student-worker) can obtain the 3 marks by taking a test at the end of the semester
  - These students, upon starting the additional test, will automatically lose any marks that were obtained from attending classes
- This component is not subject to a minimum grade

Study elements

- Android
  - "Kotlin Programming Language", https://kotlinlang.org/
  - "Android", http://www.android.com
  - "Android Developers", http://developer.android.com
  - Ricardo Queirós, 2014, "Development of Professional Applications on Android", FCA Editora
  - There are several books in the library about programming for Android
- iOS
  - "Apple Developer", http://developer.apple.com
  - "Swift – Apple Developer", https://developer.apple.com/swift/
- Flutter
  - "Flutter", https://flutter.dev
  - "Flutter Documentation", https://flutter.dev/docs
  - "Dart programming language", https://dart.dev/
  - "Dart documentation", https://dart.dev/guides
  - "DartPad", https://dartpad.dartlang.org/
  - "Dart and Flutter packages", https://pub.dev/
- Generic
  - http://www.stackoverflow.com
  - http://www.codeproject.com
- Slides are an aid…
  - for the teacher to present the content
  - for the student to organize the study, and should not be the main element of study